

A Corrected Note on Nonscalable Operator Sinkhorn Limits

Prepared for the open-problem verification request

June 23, 2026

Abstract

This note repairs what can be repaired from a proposed proof of a limit-cycle theorem for the unmodified operator Sinkhorn iteration in the nonscalable case. The proposed proof used a general principle asserting that exact alternating minimization of a lower-unbounded convex function escapes toward the unique minimizer of the recession function. That principle is false, even for smooth convex functions of logarithmic/linear growth with unique one-block minimizers and a unique recession minimizer. Consequently the proposed proof does not solve the operator-Sinkhorn open problem.

The positive results below are rigorous but partial. First, I give an explicit counterexample to the false alternating-minimization principle. Second, I prove the complete two-cycle formula for block-diagonal tuples whose diagonal blocks are exactly scalable. This verifies the algebraic limit candidate that would arise after a generalized Dulmage–Mendelsohn decoupling, and it shows that the intermediate “harmonic” block weights cancel from the actual row- and column-normalized limit points. Third, I formulate a precise conditional reduction: if the missing dynamical assertion—strict block escape and diagonal convergence for the original nonscalable operator Sinkhorn iteration—is established, then the expected two-cycle and shrunk-subspace certificate follow. Thus the note gives partial progress and isolates the remaining open step; it is not a complete solution of the authors’ problem.

1 Status and relation to the open problem

Hayashi, Hirai and Sakabe proved that, for nonscalable ordinary matrix scaling, the Sinkhorn iteration reveals Hall blockers and that the nonscalable limit is governed by Dulmage–Mendelsohn-type structure. They explicitly ask whether the analogous unmodified operator Sinkhorn iteration can reveal a Hall blocker, namely a shrunk subspace, and they point out that this may require characterizing the limits of operator Sinkhorn in the nonscalable case [4]. Hirai and Sakabe later characterized the corresponding *gradient-flow/gradient-descent* limit for operator scaling using a vector-space Dulmage–Mendelsohn decomposition [5]. Franks, Soma and Goemans gave a modified Sinkhorn-style algorithm that finds shrunk subspaces [3]. These results do not, by themselves, characterize the limit of the original unmodified alternating-normalization iteration.

The submitted manuscript attempted to bridge this gap by invoking a broad exact-alternating-minimization theorem. Section 3 proves that such a theorem is false. The rest of the note records what remains valid after removing that step.

Conclusion of this note

The results below constitute *partial progress only*. They do not prove that the unmodified operator Sinkhorn iterates of every nonscalable tuple converge to the generalized Dulmage–Mendelsohn two-cycle. They prove the block-diagonal model exactly and reduce the full conjectural statement to a sharply stated dynamical assertion about the original alternating-normalization sequence.

2 Operator Sinkhorn notation

Let

$$A = (A_1, \dots, A_k), \quad A_i \in \mathbb{C}^{M \times N}.$$

Define the two marginals

$$\text{row}(A) := \sum_{i=1}^k A_i A_i^*, \quad \text{col}(A) := \sum_{i=1}^k A_i^* A_i.$$

Throughout, inverses are taken on the full support. In the main algebraic results we assume

$$\text{row}(A) \succ 0, \quad \text{col}(A) \succ 0,$$

so no support deletion is needed. The two standard one-sided normalizations are

$$R(A) := A N^{-1/2} \text{col}(A)^{-1/2}, \quad L(A) := M^{-1/2} \text{row}(A)^{-1/2} A,$$

where multiplication of a tuple by matrices is componentwise. Thus

$$\text{col}(R(A)) = \frac{1}{N} I_N, \quad \text{row}(L(A)) = \frac{1}{M} I_M.$$

The unmodified operator Sinkhorn sequence is

$$A^{2t+1} := R(A^{2t}), \quad A^{2t+2} := L(A^{2t+1}), \quad A^0 = A. \quad (2.1)$$

For a rectangular block tuple $C = (C_i)$ with $C_i \in \mathbb{C}^{r \times c}$, write

$$R_{r,c}(C) := C C^{-1/2} \text{col}(C)^{-1/2}, \quad L_{r,c}(C) := r^{-1/2} \text{row}(C)^{-1/2} C.$$

This is the local Sinkhorn iteration with target marginals I_r/r and I_c/c .

Definition 2.1 (exact block scalability). *A tuple $C = (C_i) \in (\mathbb{C}^{r \times c})^k$ is called exactly block-scalable if there are invertible matrices $G \in GL_r(\mathbb{C})$ and $H \in GL_c(\mathbb{C})$ such that*

$$\text{row}(GCH) = \frac{1}{r} I_r, \quad \text{col}(GCH) = \frac{1}{c} I_c. \quad (2.2)$$

When the usual stabilizer is nontrivial, the scaled tuple is unique only modulo its compact unitary stabilizer; all convergence statements below may be read either after fixing the positive square-root Sinkhorn convention or modulo that compact group.

We use the standard convergence theorem for the scalable case: if C is exactly block-scalable and has full support, then its local Sinkhorn iteration converges, modulo compact unitary stabilizer, to a tuple B satisfying (2.2). This is the rectangular prescribed-marginal version of the usual operator-scaling convergence theorem; it follows, for instance, from geodesic convexity of the Kempf–Ness potential with prescribed marginals.

3 The missing alternating-minimization lemma is false

The submitted proof used the following informal principle: exact cyclic block minimization of a lower-unbounded convex function should escape to the unique minimizer of the recession function. This is false. The example below is deliberately close to the logarithmic growth behavior of Kempf–Ness potentials: it is smooth, convex, has unique exact minimizers in both blocks, is lower-unbounded, and has a unique recession minimizer.

Let

$$\varphi(s) := \log(2 \cosh s),$$

and define $F : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$F(x, y) = \varphi(x_1 - 2y_1) + \varphi(x_2 - y_2) + \frac{1}{2}x_1 + \frac{1}{2}x_2. \quad (3.1)$$

Proposition 3.1. *The function F in (3.1) has the following properties.*

- (i) F is C^∞ , convex, and lower-unbounded.
- (ii) For every fixed y , the minimizer in x is unique. For every fixed x , the minimizer in y is unique.
- (iii) The recession function of F has a unique minimizer on the Euclidean unit sphere, namely the direction proportional to

$$d_* = (-8, -5, -4, -5) \quad \text{in coordinates } (x_1, x_2, y_1, y_2). \quad (3.2)$$

- (iv) Exact alternating minimization of F escapes in the direction proportional to

$$d_{\text{AM}} = (-2, -2, -1, -2), \quad (3.3)$$

which is not proportional to d_* .

Consequently, the general alternating-minimization escape principle used in the submitted proof is invalid.

Proof. The function φ is smooth and convex, and $\varphi''(s) = \text{sech}^2(s) > 0$. Hence F is convex and is strictly convex in each block separately.

For fixed y , the first-order equations in x are

$$\tanh(x_1 - 2y_1) + \frac{1}{2} = 0, \quad \tanh(x_2 - y_2) + \frac{1}{2} = 0.$$

Thus

$$x_1 = 2y_1 - \beta, \quad x_2 = y_2 - \beta, \quad \beta := \text{artanh}(1/2) = \frac{1}{2} \log 3. \quad (3.4)$$

For fixed x , the first-order equations in y are

$$-2 \tanh(x_1 - 2y_1) = 0, \quad -\tanh(x_2 - y_2) = 0,$$

so

$$y_1 = \frac{1}{2}x_1, \quad y_2 = x_2. \quad (3.5)$$

The separate minimizers are therefore unique. Combining (3.4) and (3.5), one full two-block cycle sends

$$(y_1, y_2) \mapsto \left(y_1 - \frac{\beta}{2}, y_2 - \beta \right).$$

At the corresponding half-steps, the full vector (x_1, x_2, y_1, y_2) changes asymptotically by

$$(-\beta, -\beta, -\beta/2, -\beta),$$

which is proportional to (3.3). This proves (iv), assuming the recession minimizer calculation below.

Since $\varphi^\infty(s) = |s|$, the recession function is

$$F^\infty(u_1, u_2, v_1, v_2) = |u_1 - 2v_1| + |u_2 - v_2| + \frac{1}{2}u_1 + \frac{1}{2}u_2. \quad (3.6)$$

It is enough to minimize this positively homogeneous convex function on the unit sphere. Write it as the sum of two support functions,

$$\begin{aligned} |u_1 - 2v_1| + \frac{1}{2}u_1 &= \max \left\{ \frac{3}{2}u_1 - 2v_1, -\frac{1}{2}u_1 + 2v_1 \right\}, \\ |u_2 - v_2| + \frac{1}{2}u_2 &= \max \left\{ \frac{3}{2}u_2 - v_2, -\frac{1}{2}u_2 + v_2 \right\}. \end{aligned}$$

For a support function $h_K(z) = \max_{g \in K} \langle g, z \rangle$, the minimum of h_K on the unit sphere is $-\text{dist}(0, K)$, and the minimizing direction is the negative of the closest point of K to the origin, provided that closest point is unique and nonzero.

For the first pair (u_1, v_1) , the relevant segment has endpoints $(3/2, -2)$ and $(-1/2, 2)$. Its closest point to the origin is $(2/5, 1/5)$. For the second pair (u_2, v_2) , the relevant segment has endpoints $(3/2, -1)$ and $(-1/2, 1)$, and its closest point to the origin is $(1/4, 1/4)$. Therefore the closest point of the product convex set, written in the coordinate order (u_1, u_2, v_1, v_2) , is

$$g_* = (2/5, 1/4, 1/5, 1/4).$$

The unique minimizing recession direction is $-g_*/\|g_*\|$, which is proportional to $(-8, -5, -4, -5)$. This proves (iii). Since $F^\infty(d_*) < 0$, F is lower-unbounded. Finally, (3.2) and (3.3) are not proportional, so exact alternating minimization does not escape toward the recession minimizer. $\square$

Remark 3.2. *This counterexample is not an operator-scaling Kempf–Ness potential. It therefore does not disprove the operator-Sinkhorn conjectural picture. It does prove that the submitted manuscript’s bridge from known gradient-flow results to exact alternating normalization cannot be justified by a general theorem of convex analysis. Any successful proof must use special structure of the operator-scaling potential or a different argument.*

4 A rigorous two-cycle theorem for block-diagonal tuples

The next theorem is the main positive result of this note. It proves the expected period-two limit for the model obtained after all off-diagonal Dulmage–Mendelsohn blocks have vanished.

Fix orthogonal decompositions

$$\mathbb{C}^M = E_1 \oplus \cdots \oplus E_m, \quad \mathbb{C}^N = F_1 \oplus \cdots \oplus F_m,$$

where $\dim E_\alpha = r_\alpha$ and $\dim F_\alpha = c_\alpha$. Let

$$A = \bigoplus_{\alpha=1}^m A^{(\alpha)}, \quad A_i = \bigoplus_{\alpha=1}^m A_i^{(\alpha)}, \quad (4.1)$$

where $A^{(\alpha)} = (A_i^{(\alpha)})_{i=1}^k \in (\mathbb{C}^{r_\alpha \times c_\alpha})^k$.

Theorem 4.1 (block-diagonal operator Sinkhorn two-cycle). *Assume that every diagonal block tuple $A^{(\alpha)}$ has full support and is exactly block-scalable. For each α , let $B^{(\alpha)}$ be the local Sinkhorn limit of $A^{(\alpha)}$, so that*

$$\text{row}(B^{(\alpha)}) = \frac{1}{r_\alpha} I_{E_\alpha}, \quad \text{col}(B^{(\alpha)}) = \frac{1}{c_\alpha} I_{F_\alpha}. \quad (4.2)$$

Then the global unmodified Sinkhorn sequence (2.1) for A converges to the period-two orbit

$$\lim_{t \rightarrow \infty} A^{2t+1} = B_R := \bigoplus_{\alpha=1}^m \sqrt{\frac{c_\alpha}{N}} B^{(\alpha)}, \quad (4.3)$$

$$\lim_{t \rightarrow \infty} A^{2t+2} = B_L := \bigoplus_{\alpha=1}^m \sqrt{\frac{r_\alpha}{M}} B^{(\alpha)}. \quad (4.4)$$

Moreover

$$\text{col}(B_R) = \frac{1}{N} I_N, \quad \text{row}(B_L) = \frac{1}{M} I_M,$$

and

$$\mathbf{R}(B_L) = B_R, \quad \mathbf{L}(B_R) = B_L. \quad (4.5)$$

Proof. Because A is block diagonal, all its Sinkhorn iterates remain block diagonal. Let $C^{(\alpha),t}$ denote the local Sinkhorn sequence of $A^{(\alpha)}$ with target marginals I_{r_α}/r_α and I_{c_α}/c_α .

We claim by induction that the global and local iterates are related by

$$A_{\alpha\alpha}^{2t+1} = \sqrt{\frac{c_\alpha}{N}} C^{(\alpha),2t+1}, \quad A_{\alpha\alpha}^{2t+2} = \sqrt{\frac{r_\alpha}{M}} C^{(\alpha),2t+2}. \quad (4.6)$$

For the first right-normalization step,

$$\mathbf{R}(A^{(\alpha)}) = A^{(\alpha)} N^{-1/2} \text{col}(A^{(\alpha)})^{-1/2} = \sqrt{\frac{c_\alpha}{N}} \mathbf{R}_{r_\alpha, c_\alpha}(A^{(\alpha)}),$$

which gives the first identity at $t = 0$. Suppose the odd identity in (4.6) holds. Since multiplying a tuple by a scalar s multiplies its row marginal by s^2 ,

$$\begin{aligned} \mathbf{L}\left(\sqrt{\frac{c_\alpha}{N}} C^{(\alpha),2t+1}\right) &= M^{-1/2} \left(\frac{c_\alpha}{N} \text{row}(C^{(\alpha),2t+1})\right)^{-1/2} \sqrt{\frac{c_\alpha}{N}} C^{(\alpha),2t+1} \\ &= M^{-1/2} \text{row}(C^{(\alpha),2t+1})^{-1/2} C^{(\alpha),2t+1} \\ &= \sqrt{\frac{r_\alpha}{M}} C^{(\alpha),2t+2}. \end{aligned}$$

This proves the even identity. The implication from the even identity to the next odd identity is identical:

$$\begin{aligned} \mathbf{R}\left(\sqrt{\frac{r_\alpha}{M}} C^{(\alpha),2t+2}\right) &= N^{-1/2} C^{(\alpha),2t+2} \text{col}(C^{(\alpha),2t+2})^{-1/2} \\ &= \sqrt{\frac{c_\alpha}{N}} C^{(\alpha),2t+3}. \end{aligned}$$

Thus (4.6) holds for all t .

By the scalable-case convergence theorem, $C^{(\alpha),t} \rightarrow B^{(\alpha)}$ modulo compact unitary stabilizer. Passing to the limit in (4.6) proves (4.3) and (4.4). The marginal identities are immediate from (4.2):

$$\begin{aligned} \text{col} \left(\sqrt{\frac{c_\alpha}{N}} B^{(\alpha)} \right) &= \frac{c_\alpha}{N} \frac{1}{c_\alpha} I_{F_\alpha} = \frac{1}{N} I_{F_\alpha}, \\ \text{row} \left(\sqrt{\frac{r_\alpha}{M}} B^{(\alpha)} \right) &= \frac{r_\alpha}{M} \frac{1}{r_\alpha} I_{E_\alpha} = \frac{1}{M} I_{E_\alpha}. \end{aligned}$$

Finally,

$$\text{col} \left(\sqrt{\frac{r_\alpha}{M}} B^{(\alpha)} \right) = \frac{r_\alpha}{M c_\alpha} I_{F_\alpha}.$$

Right-normalizing this block multiplies it by

$$N^{-1/2} \left(\frac{r_\alpha}{M c_\alpha} I_{F_\alpha} \right)^{-1/2} = \sqrt{\frac{M c_\alpha}{N r_\alpha}} I_{F_\alpha},$$

which sends $\sqrt{r_\alpha/M} B^{(\alpha)}$ to $\sqrt{c_\alpha/N} B^{(\alpha)}$. Hence $R(B_L) = B_R$. The proof of $L(B_R) = B_L$ is the same with rows and columns interchanged. $\square$

4.1 The intermediate block weights cancel

The submitted manuscript introduced weights

$$\lambda_\alpha = \frac{c_\alpha}{C(r_\alpha + c_\alpha)}, \quad \rho_\alpha = \frac{r_\alpha}{C(r_\alpha + c_\alpha)}. \quad (4.7)$$

Those weights may be natural for a particular boundary or moment-polytope calculation. However, for the actual normalized two-cycle, the value of the intermediate mass is irrelevant.

Proposition 4.2 (mass cancellation). *Let $\mu_1, \dots, \mu_m > 0$ with $\sum_\alpha \mu_\alpha = 1$, and let*

$$\tilde{B} := \bigoplus_{\alpha=1}^m \sqrt{\mu_\alpha} B^{(\alpha)}.$$

Then

$$\text{row}(\tilde{B})|_{E_\alpha} = \frac{\mu_\alpha}{r_\alpha} I_{E_\alpha}, \quad \text{col}(\tilde{B})|_{F_\alpha} = \frac{\mu_\alpha}{c_\alpha} I_{F_\alpha}.$$

The two one-sided normalizations of $\tilde{B}$ are exactly the two tuples in (4.3) and (4.4), independently of the chosen masses μ_α :

$$R(L(\tilde{B})) = B_R, \quad L(R(\tilde{B})) = B_L,$$

and directly

$$M^{-1/2} \text{row}(\tilde{B})^{-1/2} \tilde{B} = B_L, \quad \tilde{B} N^{-1/2} \text{col}(\tilde{B})^{-1/2} = B_R. \quad (4.8)$$

Proof. The marginal formula follows from (4.2). For the left normalization on block α ,

$$M^{-1/2} \left(\frac{\mu_\alpha}{r_\alpha} I_{E_\alpha} \right)^{-1/2} \sqrt{\mu_\alpha} B^{(\alpha)} = \sqrt{\frac{r_\alpha}{M}} B^{(\alpha)}.$$

The right normalization is identical:

$$\sqrt{\mu_\alpha} B^{(\alpha)} N^{-1/2} \left(\frac{\mu_\alpha}{c_\alpha} I_{F_\alpha} \right)^{-1/2} = \sqrt{\frac{c_\alpha}{N}} B^{(\alpha)}.$$

Thus μ_α cancels. $\square$

Remark 4.3. *This does not say that the harmonic weights are wrong as a description of a possible “middle” scaled tuple. It says that, after imposing the actual one-sided normalizations used by the unmodified Sinkhorn iteration, every positive mass vector produces the same two limit points. Therefore a proof of the normalized limit cycle should not depend on uniqueness of the harmonic masses unless it is also proving a stronger statement about the cumulative scaling matrices.*

5 A conditional reduction for the full nonscalable problem

The open problem concerns tuples that are not block diagonal. The expected picture is that a generalized Dulmage–Mendelsohn flag makes the tuple block upper triangular, that the off-diagonal blocks decay under Sinkhorn, and that the diagonal blocks converge to the block-scalable model of Section 4. The next result gives a clean reduction: once those dynamical assertions are proved, the two-cycle follows with no further hidden argument.

Fix block decompositions $\mathbb{C}^M = \bigoplus_{\alpha} E_{\alpha}$ and $\mathbb{C}^N = \bigoplus_{\alpha} F_{\alpha}$ with dimensions r_{α}, c_{α} . Write Π_{diag} for the projection of a tuple onto the diagonal blocks and $\Pi_{\text{off}} := I - \Pi_{\text{diag}}$.

Hypothesis 5.1 (strict decoupling and diagonal convergence). *For the Sinkhorn sequence (A^t) in (2.1), suppose that:*

(H1) $\|\Pi_{\text{off}} A^t\| \rightarrow 0$.

(H2) *For each diagonal block there is an exactly block-scalable tuple $B^{(\alpha)}$ satisfying (4.2) such that*

$$A_{\alpha\alpha}^{2t+1} \rightarrow \sqrt{\frac{c_{\alpha}}{N}} B^{(\alpha)}, \quad A_{\alpha\alpha}^{2t+2} \rightarrow \sqrt{\frac{r_{\alpha}}{M}} B^{(\alpha)}. \quad (5.1)$$

Proposition 5.2 (formal consequence of decoupling). *Under Hypothesis 5.1, the full Sinkhorn sequence converges to the two-cycle*

$$A^{2t+1} \rightarrow B_R := \bigoplus_{\alpha} \sqrt{\frac{c_{\alpha}}{N}} B^{(\alpha)}, \quad A^{2t+2} \rightarrow B_L := \bigoplus_{\alpha} \sqrt{\frac{r_{\alpha}}{M}} B^{(\alpha)}.$$

The tuples B_R and B_L satisfy

$$\text{col}(B_R) = \frac{1}{N} I_N, \quad \text{row}(B_L) = \frac{1}{M} I_M, \quad R(B_L) = B_R, \quad L(B_R) = B_L. \quad (5.2)$$

Proof. The convergence statements follow by adding the convergent diagonal blocks in (5.1) and the vanishing off-diagonal part. The identities (5.2) are exactly the identities proved in Theorem 4.1. $\square$

The nontrivial part is therefore not the two-cycle algebra; it is proving Hypothesis 5.1 for the generalized Dulmage–Mendelsohn block decomposition of an arbitrary nonscalable tuple. The following elementary lemma records one standard way in which the first half of this hypothesis would follow from a strict escape statement for the cumulative scaling matrices.

Lemma 5.3 (strict block escape implies off-diagonal decay). *Let A be written in blocks $A_{\alpha\beta} : F_{\beta} \rightarrow E_{\alpha}$. Suppose a sequence of scaled tuples has the form*

$$Z_i^t = G_t A_i H_t,$$

where, for some $\tau_t \rightarrow \infty$,

$$G_t = U_t \left(\bigoplus_{\alpha} e^{\tau_t p_{\alpha}} I_{E_{\alpha}} \right), \quad H_t = \left(\bigoplus_{\beta} e^{\tau_t q_{\beta}} I_{F_{\beta}} \right) V_t, \quad (5.3)$$

with U_t, V_t uniformly bounded in operator norm. Assume that

$$p_{\alpha} + q_{\alpha} = 0 \quad \text{for all diagonal blocks,} \quad p_{\alpha} + q_{\beta} < 0 \quad \text{whenever } \alpha \neq \beta \text{ and } A_{\alpha\beta} \neq 0. \quad (5.4)$$

Then every off-diagonal block of Z^t converges to zero. If the diagonal block sequences

$$U_{t,\alpha} A_{\alpha\alpha} V_{t,\alpha}$$

converge to block-scaled limits, then Z^t converges to the direct sum of those limits.

Proof. The (α, β) block of Z_i^t is

$$e^{\tau_t(p_{\alpha} + q_{\beta})} U_{t,\alpha} A_{i,\alpha\beta} V_{t,\beta}.$$

For $\alpha \neq \beta$, the exponent in (5.4) is strictly negative, while U_t, V_t are uniformly bounded. Hence this block tends to zero. For $\alpha = \beta$, the exponent is zero, so convergence is exactly the assumed convergence of the bounded internal scaling sequence. $\square$

Remark 5.4. For gradient flow/descent of the Kempf–Ness potential, the analog of the escape direction is supplied by the minimum-norm point of the moment polytope and, in the operator-scaling case, by the generalized Dulmage–Mendelsohn decomposition [5]. Proposition 3.1 shows that transferring this conclusion to exact alternating minimization is not automatic. Establishing the analog of (5.3)–(5.4) for the actual cumulative matrices generated by unmodified operator Sinkhorn remains the essential missing theorem.

6 Shrunk-subspace certificate once the flag is known

The certificate part of the expected theorem is elementary. It does not require any convergence of Sinkhorn; it only requires an independent pair.

Assume now $M = N = n$. For a tuple $A = (A_i)$, call a pair of subspaces (X, Y) , with $X, Y \leq \mathbb{C}^n$, independent if

$$x^* A_i y = 0 \quad \text{for all } x \in X, y \in Y, \text{ and all } i. \quad (6.1)$$

Equivalently, $A_i Y \subseteq X^{\perp}$ for every i .

Proposition 6.1 (independent-pair certificate). *If (X, Y) is independent and*

$$\dim X + \dim Y > n, \quad (6.2)$$

then Y is a shrunk subspace:

$$\dim \left(\sum_i A_i Y \right) < \dim Y. \quad (6.3)$$

Proof. Independence gives $\sum_i A_i Y \subseteq X^{\perp}$. Therefore

$$\dim \left(\sum_i A_i Y \right) \leq \dim X^{\perp} = n - \dim X < \dim Y,$$

where the strict inequality is (6.2). $\square$

Thus, if the generalized Dulmage–Mendelsohn flag can be recovered from the limiting Sinkhorn scaling matrices and one of its breakpoints violates $\dim X + \dim Y \leq n$, the corresponding Y is automatically a shrunk subspace. What remains open for the unmodified algorithm is the recovery of such a flag from its actual limiting behavior.

7 What has been fixed, and what remains open

The attempted proof can be repaired only to the following extent.

1. The broad alternating-minimization lemma must be deleted. Proposition 3.1 gives a smooth convex counterexample with unique one-block minimizers and unique recession minimizer.
2. The two-cycle algebra is correct once diagonal block decoupling and blockwise scalability are known. Theorem 4.1 proves this completely for block-diagonal tuples.
3. The harmonic intermediate masses are not needed to identify the actual one-sided normalized limit points; Proposition 4.2 proves that they cancel.
4. The full operator-Sinkhorn open problem is reduced to proving strict decoupling and diagonal convergence, as stated in Hypothesis 5.1 or, more structurally, to proving a strict cumulative-scaling escape statement of the kind used in Lemma 5.3.

Accordingly, this note is not a solution of the open problem. It is rigorous partial progress: it verifies the candidate limit in the exactly decoupled model, identifies an incorrect proof mechanism, and isolates the precise dynamical input still needed for the nonscalable unmodified operator Sinkhorn iteration.

References

- [1] L. Gurvits. Classical complexity and quantum entanglement. *Journal of Computer and System Sciences*, 69(3):448–484, 2004.
- [2] A. Garg, L. Gurvits, R. Oliveira, and A. Wigderson. Operator scaling: theory and applications. *Foundations of Computational Mathematics*, 20:223–290, 2020.
- [3] C. Franks, T. Soma, and M. X. Goemans. Shrunk subspaces via operator Sinkhorn iteration. In *Proceedings of the 2023 Annual ACM–SIAM Symposium on Discrete Algorithms (SODA)*, pages 1655–1668, 2023.
- [4] K. Hayashi, H. Hirai, and K. Sakabe. Finding Hall blockers by matrix scaling. *Mathematics of Operations Research*, 49(4):2166–2179, 2024.
- [5] H. Hirai and K. Sakabe. Gradient descent for unbounded convex functions on Hadamard manifolds and its applications to scaling problems. arXiv:2404.09746; conference version in *FOCS 2024*. Journal version published online in *Mathematics of Operations Research*, 2026.
- [6] M. Bačák. *Convex Analysis and Optimization in Hadamard Spaces*. De Gruyter, 2014.